SOFTWARE DEVELOPMENT – 2

Sprint 4 – Final Product

Project Title

Skill Streak-A Smart Knowledge-Sharing and Skill Development Platform

Group Members:

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Introduction:

In this proposal represents the completion of our project, where all planned features are implemented from earlier sprints have been fully developed, tested, and integrated into a working web application. The goal of this sprint was to deliver a complete, stable, and deployable system that meets the objectives defined during the sprint-1, sprint-2, and sprint-3.

At the stage, the application is no longer a prototype. Its fully functional platform that allows the user to register, log in explore course, learn through structured lessons, track their progress, and interact with different system features such as reflections, leaderboards and challenges.

The system demonstrates a clear connection between design, development, and user needs.

Final Application Overview:

The developed platform is learning and knowledge-sharing system which is formally named as **SKILL STREAK** which supports both technical and non-technical skill development.

The users can:

- Create accounts and securely log in (including Google authentication)
- Browse structured courses (Basic → Intermediate → Advanced)
- Access lessons and learning materials
- Write reflections after each lesson
- Track daily learning activity using streaks
- Participate in challenges and contests
- View leaderboard rankings
- Explore job opportunities after completing courses
- Access certificates after completion of course

The platform focuses on an active learning, where users not only consume content but also engage with it through writing, practice, and consistency.

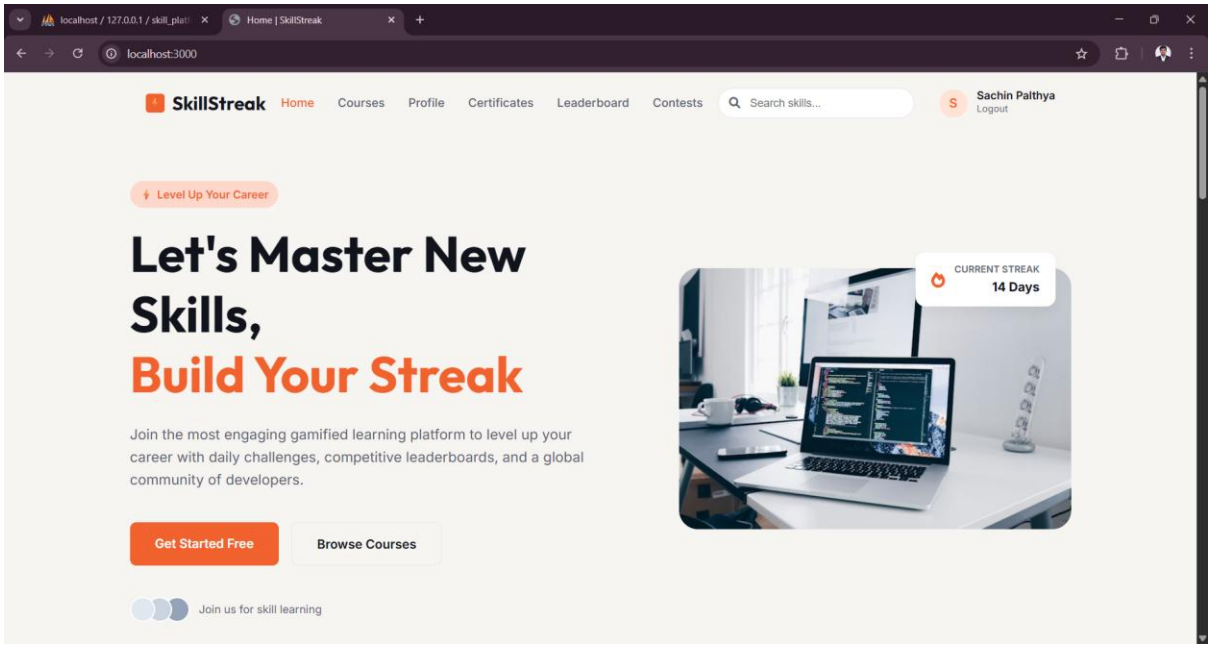

Figure: the above figures is the home page and nav bar with all pages

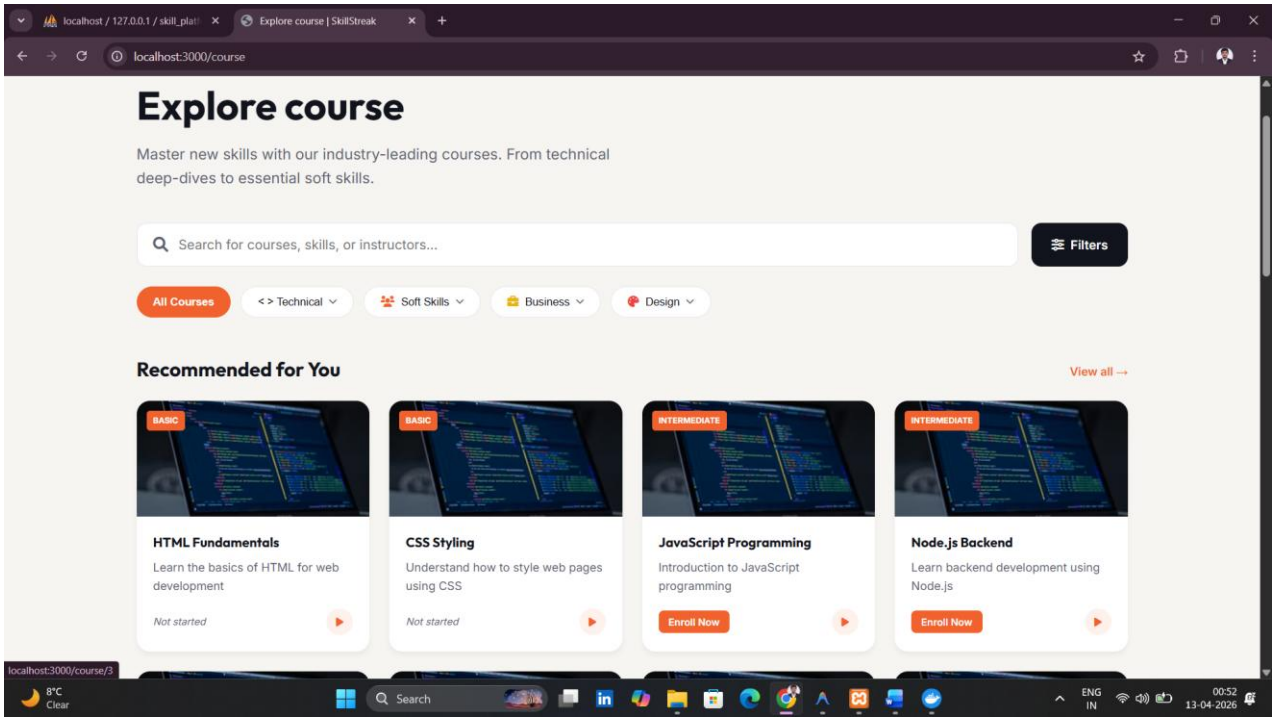

Figure: this is the course page and lesson page

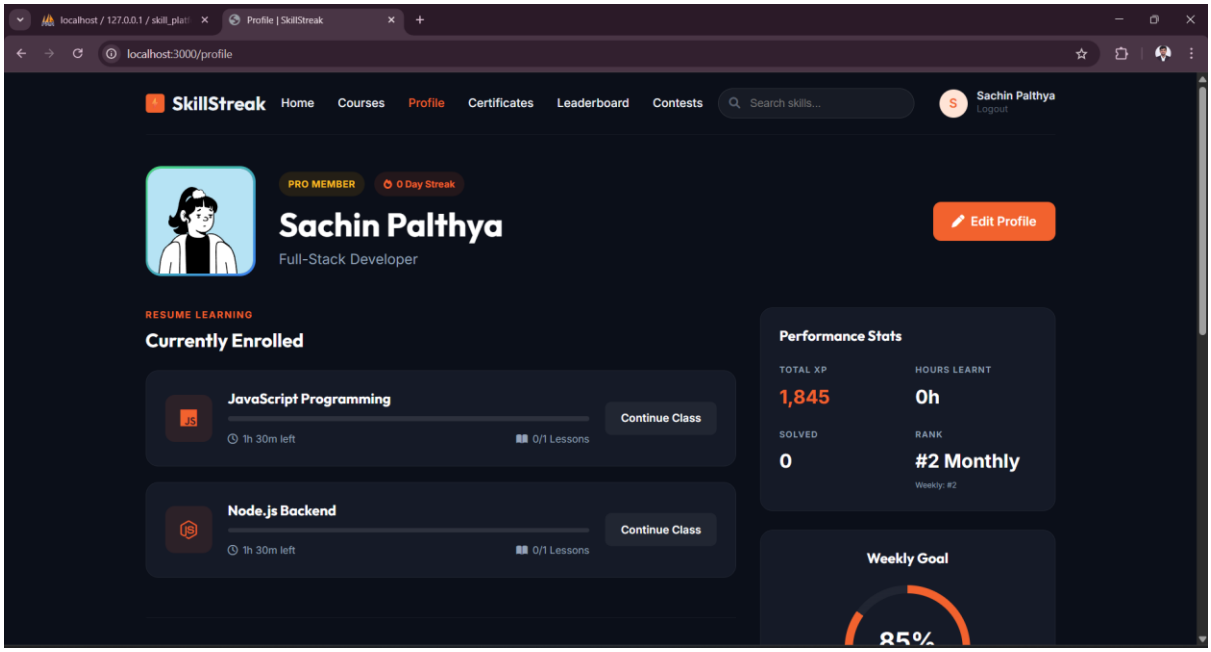

Figure: The above figure displays the profile page

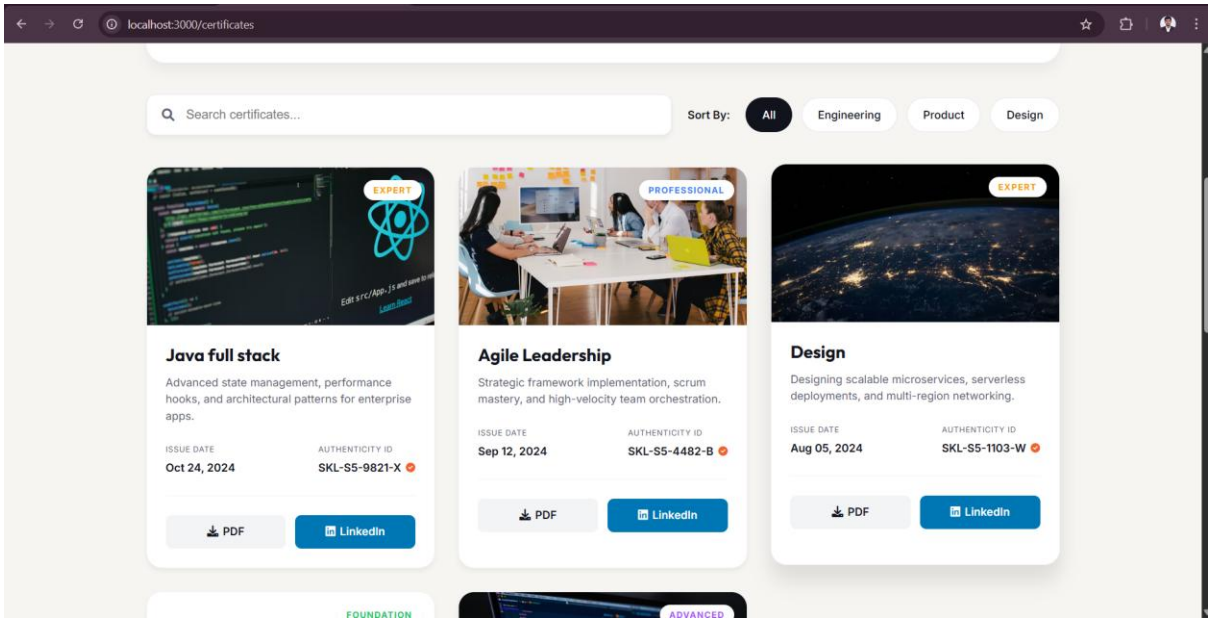

Figure: above picture shows the certificate page

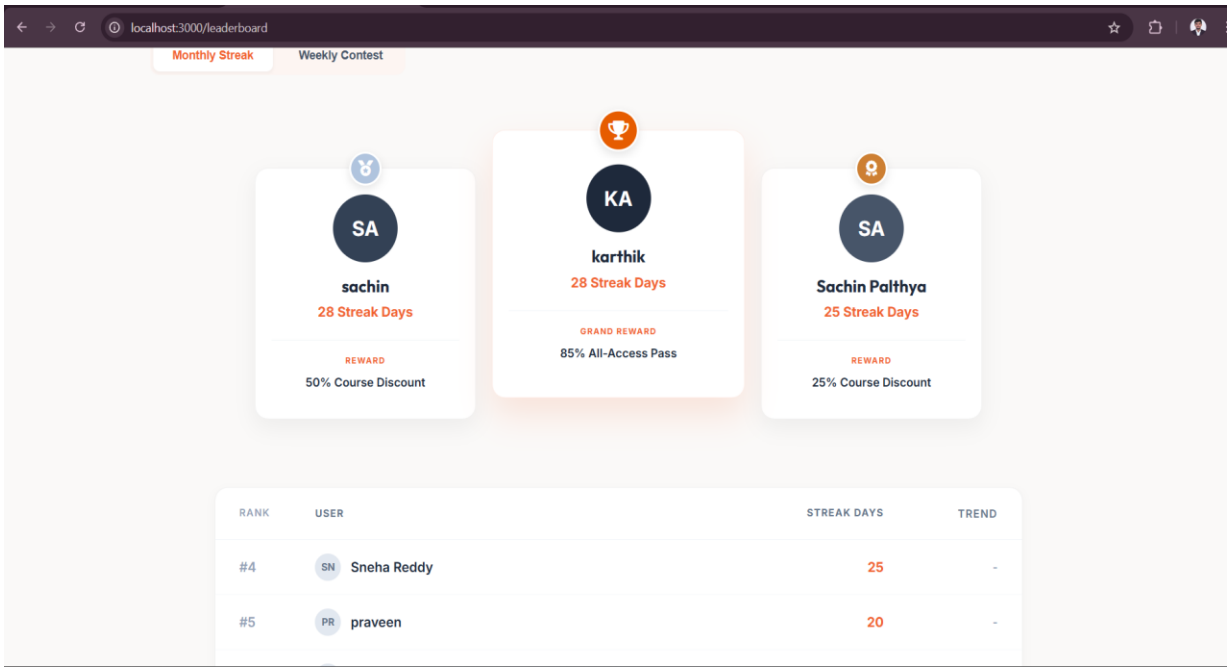

Figure: above picture shows the weekly contest and monthly streak Leaderboard

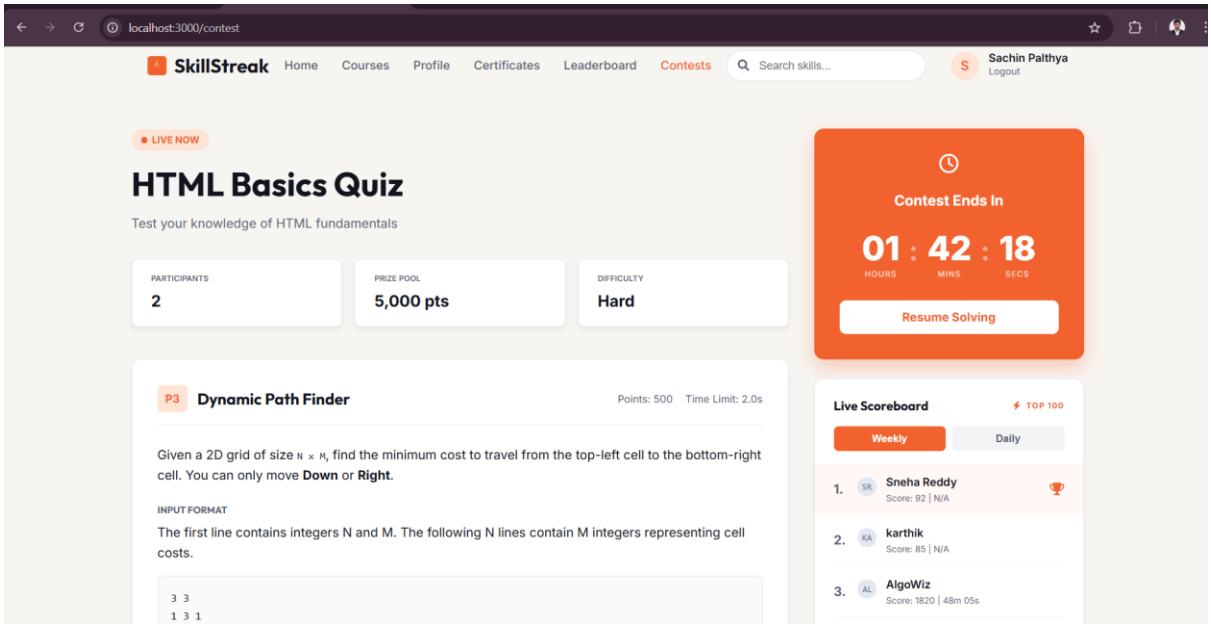

Figure: the above picture shows the context page with live context

Application Features (Fully Implemented):

Core features we implemented

- Create accounts and securely log in (including Google authentication)
- Browse structured courses (Basic → Intermediate → Advanced)
- Access lessons and learning materials
- Write reflections after each lesson
- Track daily learning activity using streaks
- Participate in challenges and contests
- View leaderboard rankings
- Explore job opportunities after completing courses

Learning Features we implemented

- Step-by-step course progression
- Level unlocking system
- Daily streak tracking
- Monthly progress monitoring

Platform Engagement Features we implemented

- Daily challenges
- Contest participation system
- Leaderboards (Streak-based and Performance-based)
- Monthly achievers display

Additional Features

- Job listings linked to completed courses
- Comment section for discussions
- Soft skills learning (**Reels – planned for future**)
- Certificate generation system (basic implementation)

User Stories Implemented in Sprint 4:

Registration Feature					
User Story ID	**Role**	**Feature**	**Reason**	**Assigned To**	**Status**
US-01	Student	Register using email and password	User registration functionality implemented in the system	Sachin Palthya	DONE
US-02	Mentor	Register using email and password	Mentor registration functionality implemented	Sachin Palthya	DONE
US-03	admin	Credentials for Admin access	Given separate admin credentials for admin support	Sachin Palthya	DONE
US-04	Student,mentor	Sign Up with google authentication	Google authentication implemented for register	Sachin palthya	Done
US-05	user	Need to show the progress of the students enrolled	Displayed the cards with student progress	Sachin Palthya	Done
Login Feature					
US-06	Student	Login using email and password	Login using email and password	Sachin palthya	DONE
US-07	USER	Need the Sign in with google	Given the option for sign in with google auth	Sachin palthya	DONE

Homepage Display					
US-08	Student	View available courses through explore course	Homepage reads and displays course data from the database	Sachin palthya	DONE
US-09	student	Need the contest challenge navigation from home page to contest	Implemented given the contest section on home page	Saraswathi velupula	DONE
US-10	user	Give certifications navigation in home page	Given the navigation for certificates	Anirudh	DONE
US-11	USER	Display the wall of frame for top scorers	Displayed the top performers on home page and navigation to leaderboard page	Sneha Paruchuri	DONE
US-12	STUDENT	Course enrollment process	The course enrollment had course section on home page and navigated to explore course	Sampath Perumula	DONE
Course Page					
US-13	Student	View courses divided into levels	Course page displays courses categorized by levels	Sachin Palthya	DONE
US-14	User	Show the jobs in the course page	Displayed the jobs in the course page	Sachin Palthya	DONE
Lessons Page					
US-15	Mentor / Student	Browse lessons in a course	Lessons page retrieves lesson data from the database	Sampath Perumula	DONE

US-16	user	Added discussions and peer explanations	Implemented the discussion and peer explanations below the lessons	Sampath Perumula	DONE
Certificate Page					
US-17	Student	Show the certificate in the categorized	Certificates displayed in the 3-4 categories	Anirudh Sidda	DONE
US-18	User	Give option for download and share to LinkedIn	Given the source for the download as pdf and share to LinkedIn in certificate page	Anirudh Sidda	DONE
Leaderboard Page					
US-19	Student	Give the leaderboard in two categories	Implemented the leaderboard in monthly streak and weekly contest leaderboard	Sneha Paruchuri	DONE
US-20	user	Show top 3 achievers in top of the leaderboard	Displayed top 3 achievers on leaderboard	Sneha Paruchuri	DONE
Profile Page					
US-21	User	Display the Enrolled Courses in the profile	Display the enrolled courses in the grid Type	Sachin Palthya	DONE
US-22	student	Add achievement gallery on the profile to make the user more connected	Implemented the achievement section and weekly gools and badges for motivation	Sachin Palthya	DONE
Contest Page					
US-23	student	Give the live contest on the top with timer	Implemented the contest with the live timer	Saraswathi velupula	DONE

US-24	USER	Give live leaderboard on the side bar for the motivation for task	Implemented the live leaderboard on the side bar with the progress	Saraswathi velupula	DONE
US-25	Mentor & Admin	Give the flow for the context in the database to update the database of contest	Provided the table contest of update and renew the tasks every day	Saraswathi Velupula	DONE
Course_Enrollment Page					
US-26	User	I want to enroll from the recommended courses	I have implemented the courses which is attached in the course page we need to enroll there	Sampath Perumula	DONE
US-27	Student	Show the enrolled courses in the profile page no separate page is required	Implemented the course page in the profile pages shown with cards	Sampath Perumula	DONE
US-28	Admin	Menton enrolled date and discount for the top leaders on the leaderboard	Designed the discount for top scorer and mentioned enrolled date in database	Sampath Perumula	DONE

Application Quality and Performance:

The final application has been developed with focus on easy to Use and make more concentrated and attached to the platform.

- Smooth navigation between pages
- Clean and simple user interface
- Proper handling of user inputs
- Error handling for invalid actions
- Consistent data display from database

All major features work correctly without system crashes. The application provides a stable experience when performing actions such as login, course access, lesson viewing, perfect leaderboards with the weekly contest and monthly streaks maintained.

The streaks and contest leaderboard are renewed every month for the month top rankers provided the offer for the intermediate and advance courses to enrol.

This ensures a high-quality user experience, which meets the expectations for an excellent-level submission.

Code Quality (MVC and OOP Design):

The project follows a structured and modular development approach based on the scaffolding architecture provided during the module. Although the folder naming differs slightly from traditional MVC, the overall design clearly follows the core principles of Model-View-Controller.

MVC Architecture Implementation

The application structure is organised as follows:

- **Model (Data Layer):**

  The models folder inside the app directory is responsible for handling all database interactions. It communicates with the MySQL database through query functions defined in the database configuration. Each model represents a specific entity such as users, courses, or lessons.

- **View (Presentation Layer):**

  The views folder contains all PUG template files used to render the user interface. These templates dynamically display data received from the backend, allowing users to interact with the system through pages such as login, courses, and lessons.

- **Controller / Service Layer (Logic Handling):**

  Instead of a traditional controller folder, the project uses a services layer along with route handling. The services folder contains the main application logic, processing user requests, validating data, and coordinating between models and views. The routes folder defines application endpoints and connects HTTP requests to the appropriate service functions.

OOP and Modular Design Principles

The application is developed using key Object-Oriented and modular programming principles:

- **Separation of Concerns:**
  Each part of the application (models, services, views) has a clearly defined responsibility, reducing complexity and improving maintainability.
- **Reusable Components:**

  Common functionalities such as database access and authentication are structured in reusable modules.
- **Modular File Organisation:**

  The project is divided into logical directories (app, routes, db, config), making it easier to navigate and extend.
- **Encapsulation:**

  Business logic is contained within service files, preventing direct access from other layers.

Best Practices Implemented

- Consistent naming conventions across files and functions
- Clear folder structure based on scaffolding design
- Use of environment variables (.env) for sensitive data
- Input validation and error handling in service layer
- Separation of static files (static) from application logic
- Database connection handled centrally (db/index.js)

GitHub Contribution and Collaboration:

Our team used GitHub regularly in all sprints to manage and track the project work.

Every team member has contributed to the project by making commits. The commits were done frequently, not all at once, which shows continuous development. Each commit includes clear messages, so it is easy to understand what changes were made.

We followed a proper teamwork approach. Tasks were divided based on user stories, and each member was responsible for specific parts of the project such as frontend, backend, or database. We combined our work step by step and made sure everything worked together properly.

We also discussed progress regularly and reviewed each other's work to avoid mistakes. This helped us improve the quality of the application.

Evidence of Contribution:

- Multiple commits from all team members
- Regular updates and improvements
- Feature-based commits (not bulk uploads)
- Clear commit messages

Collaboration Approach:

- Tasks divided using user stories
- Each member responsible for specific modules
- Continuous integration of features
- Regular discussion and review of progress

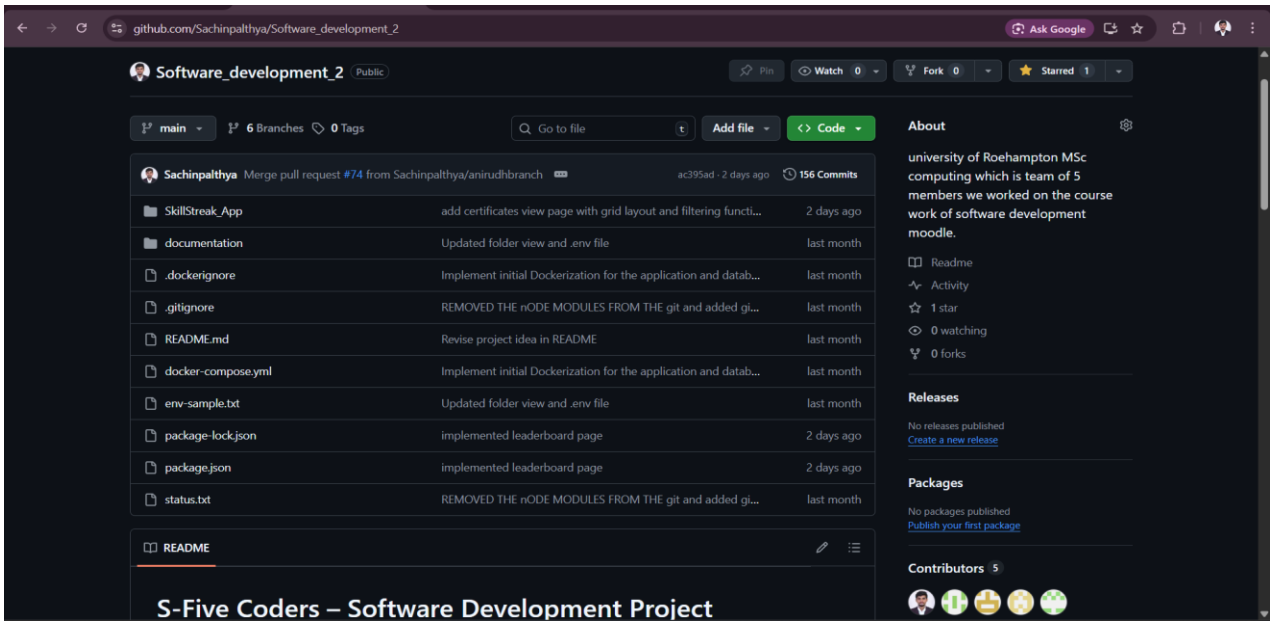

Figure: Shows the GitHub Overall structure and contributions

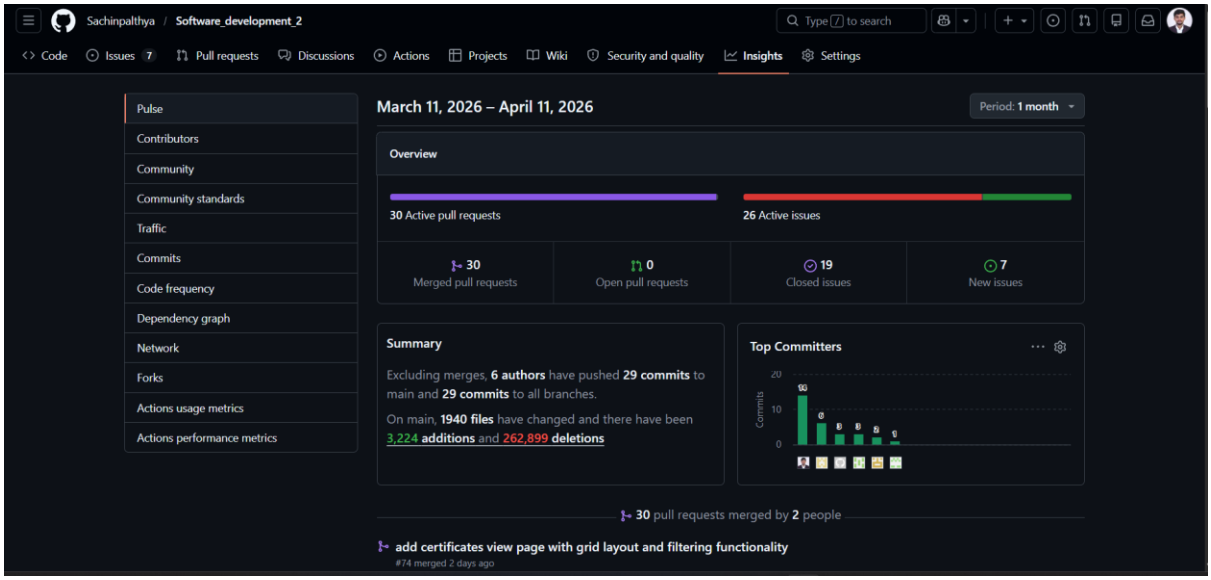

Figure: Shows the pull request done from the all-specified branches

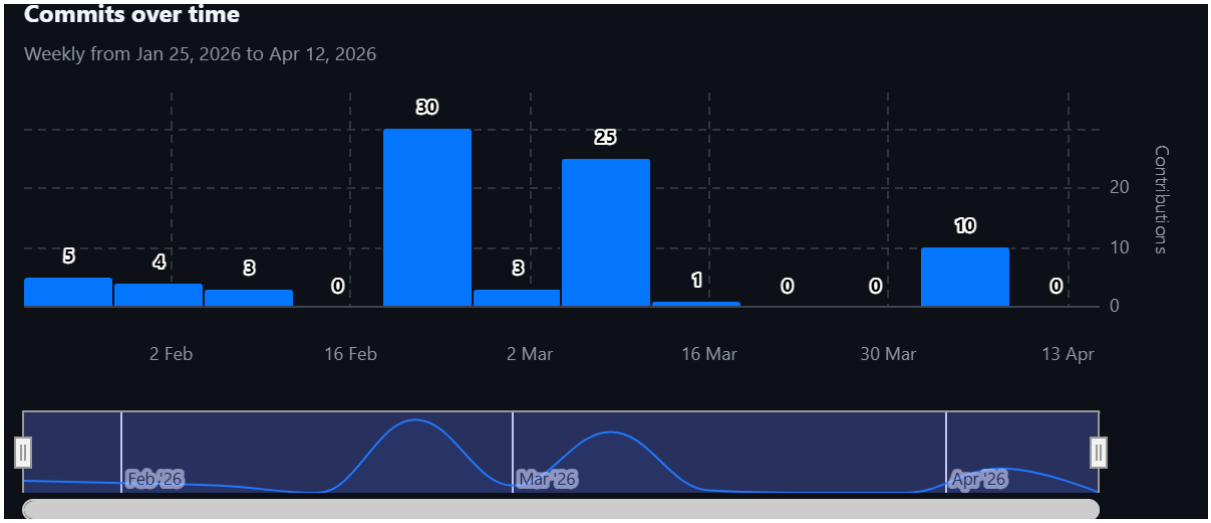

Figure : The graph shows the commits of the team over the time

Figure: it shows the team contributions made by the team members through out the project
Note: Saraswathivelpula200-tech is also made more them 14 commits due to git error the commits where not displayed we apologizes for that inconvenience. We have proofs for that in the commit messages.

Repository Link:

https://github.com/Sachinpalthya/Software_development_2.git

Task Board (Final State):

The task board reflects the full lifecycle of the project:

- Todo → Initial ideas and features
- In Progress → Features under development
- Testing → Completed features under checking
- Done → Fully implemented features

TASK BOARD LINK:

https://github.com/users/Sachinpalthya/projects/6

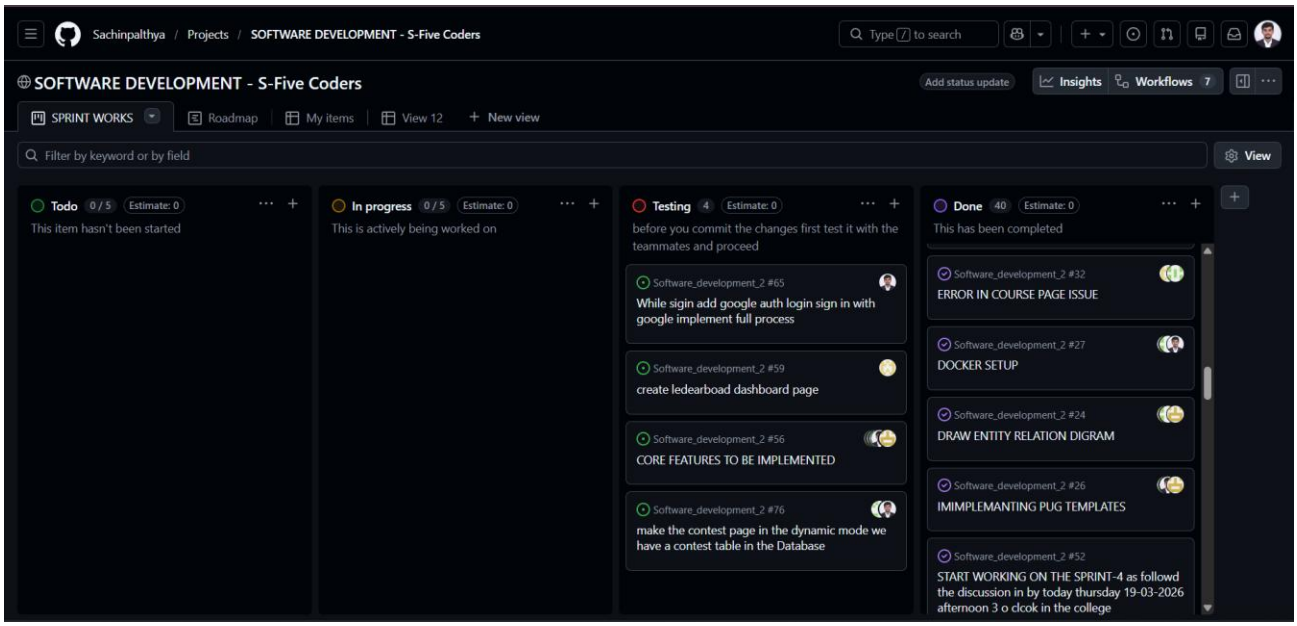

Figure: Team collaboration and task is shown in the task board

Figure: Team collaboration and task graph insights

Deployment using Docker:

The application is fully containerised using Docker.

Docker Setup Includes:

- Node.js application container
- MySQL database container
- Docker Compose configuration

docker-compose.yml manages:

- Backend service
- Database service
- Port mapping
- Environment variables

This allows the application to be run easily on any system using:

docker-compose up

The deployment setup ensures:

- Consistency across environments
- Easy installation and execution
- Professional deployment practice

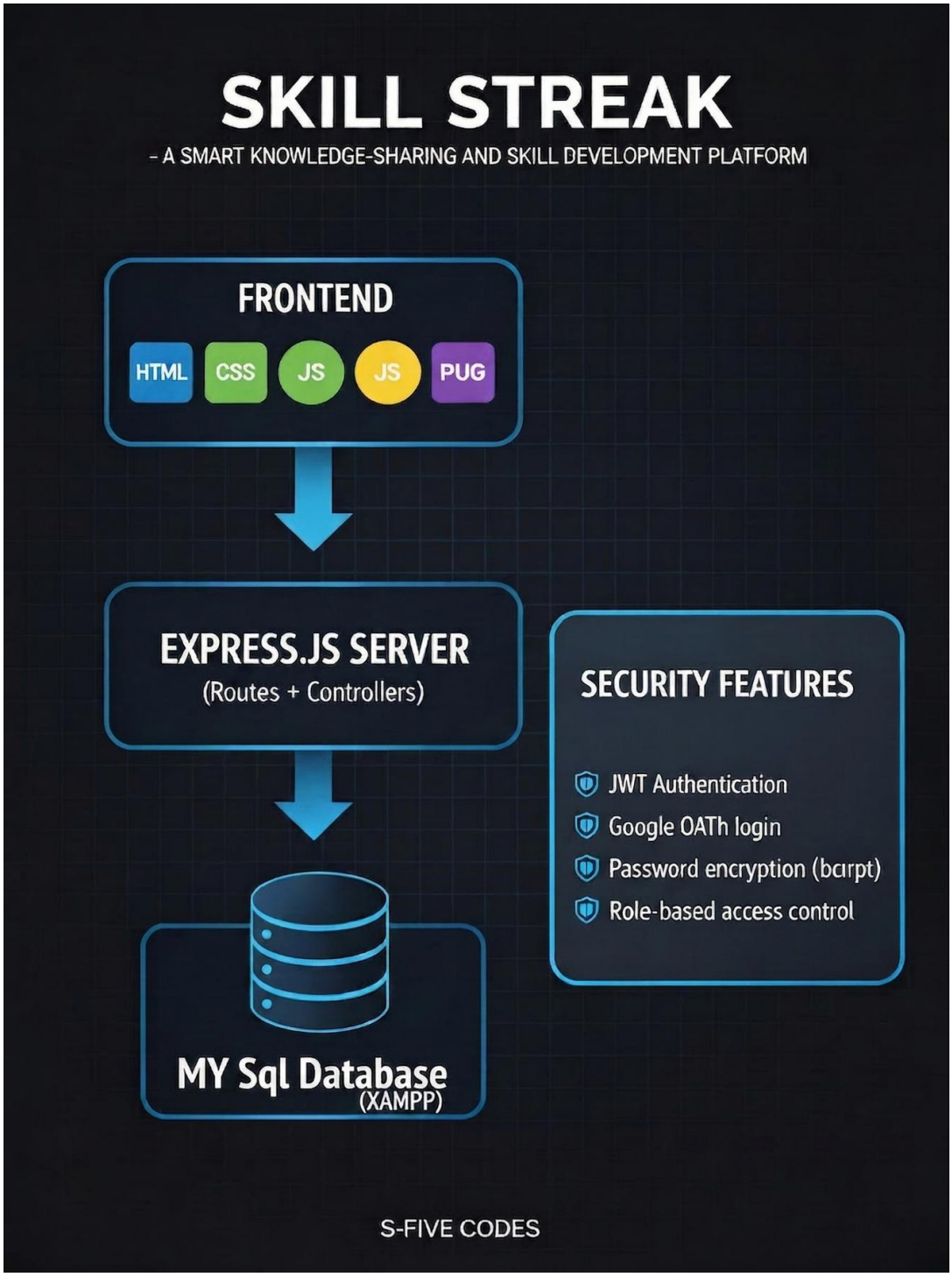

FIGURE: shows the TECH-STACK we used throughout the development of this platform

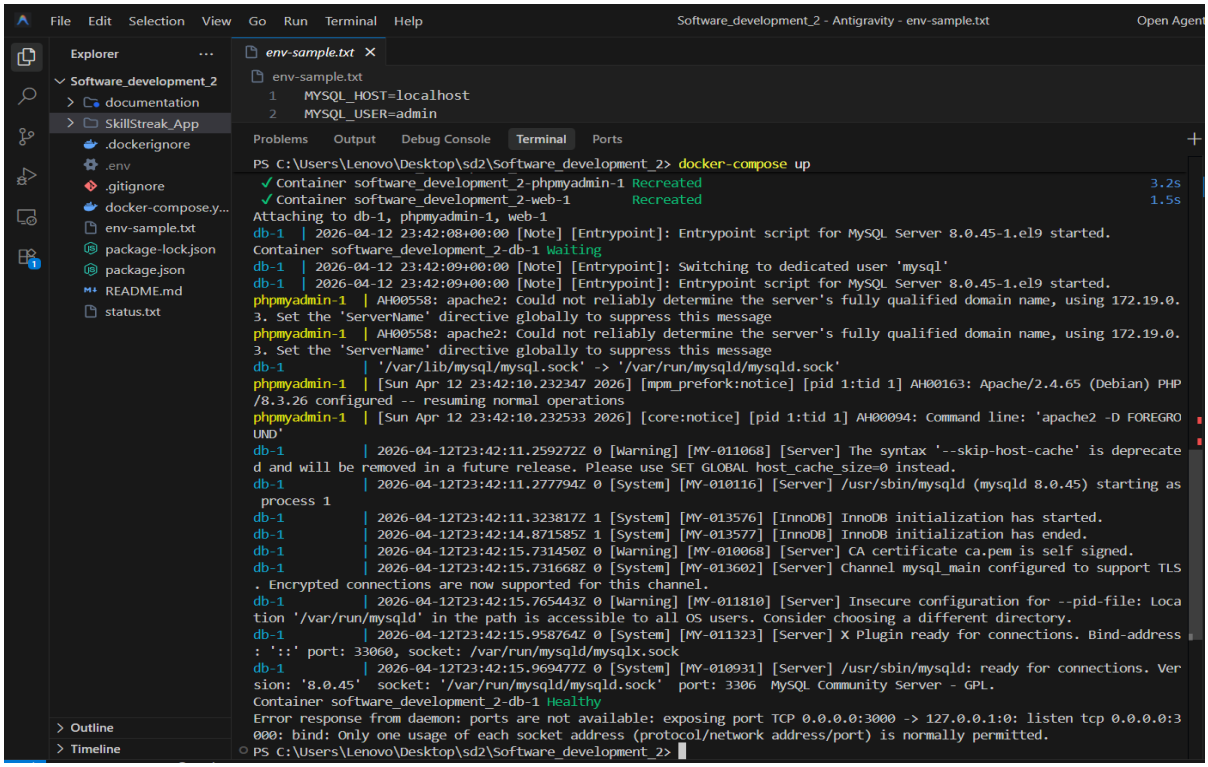

Figure: shows that the codebase is deployed using DOCKER-COMPOSE UP command

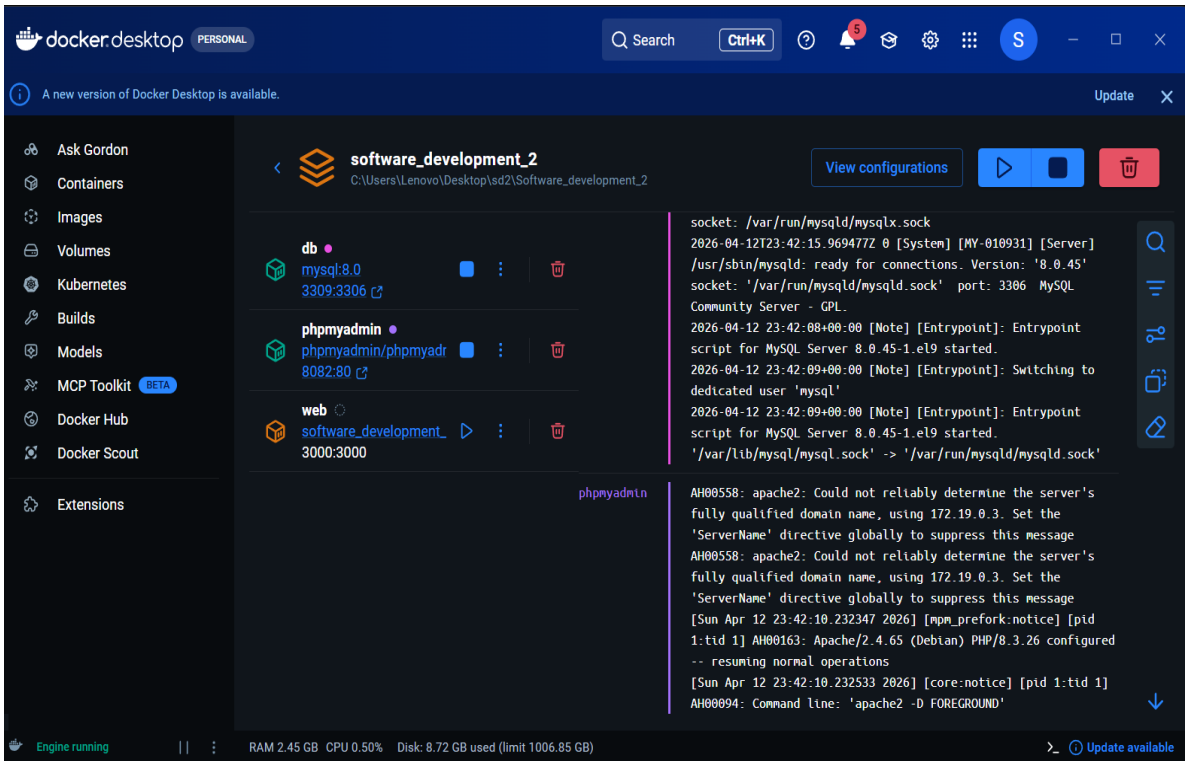

Figure: Shows that the code-base is deployed in the docker and the container are running.

Conclusion

Sprint 4 completes the full development cycle of the Smart Knowledge Sharing and Skill Development Platform. The project has successfully evolved from an initial idea into a fully functional web application that meets both user needs and technical requirements.

The final system demonstrates strong alignment between design planning and implementation, with all major features working together in a structured and meaningful way. The use of MVC architecture, secure authentication methods, and a well-organised database design reflects a solid understanding of modern web development practices.

In addition, consistent team collaboration, proper task tracking, and effective use of GitHub highlight the group's ability to work professionally in a team environment. The inclusion of Docker deployment further strengthens the project by ensuring portability and ease of use.

Overall, the application delivers a complete learning platform experience and meets the expectations of an excellent-level submission in terms of functionality, code quality, teamwork, and system design.